UNIVERSITY OF THE PEOPLE

MATH 1280-001-AY2026-T4

Assignment Activity Unit 8

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Date: 28-May-2026

Central Limit Theorem: A Case Study on Tourism in Pushkar, India

1. The Central Limit Theorem and Associated Concepts

The **Central Limit Theorem (CLT)** is one of the most foundational principles in statistics. It states that when sufficiently large random samples of size n are drawn from any population, regardless of its underlying distribution, the sampling distribution of the sample mean approaches a **normal distribution** as n increases. In practice, a sample size of $n \geq 30$ is generally considered sufficient for the CLT to apply (OpenStax, 2022). Formally, if a population has mean μ and standard deviation σ, then the sampling distribution of the sample mean $\bar{X}$ is:

$$\bar{X} \sim N(\mu, \ \sigma / \sqrt{n})$$

The **Law of Large Numbers (LLN)** is a complementary theorem stating that as the sample size increases, the sample mean converges toward the true population mean μ. While the CLT describes the *shape* of the sampling distribution, the LLN assures us of its *center*. Together, they justify the use of sample-based inference in economics, health sciences, and engineering (Triola, 2018).

The **sampling distribution of sample means** refers to the probability distribution formed by the means of all possible samples of a fixed size drawn from a population. Its standard deviation called the **standard error,** is $\sigma_{\bar{x}} = \sigma / \sqrt{n}$, which decreases as *n* grows, reflecting that larger samples produce more stable, reliable estimates.

In the context of Pushkar's tourism economy, daily revenue is recorded with a population mean of **μ = 3.7 million INR** and a standard deviation of **σ = 0.52 million INR**. Revenue may vary due to seasonal festivals, tourist influxes, and economic conditions. By applying the CLT to a random sample of 30 days, analysts can model the average daily revenue as approximately normally distributed. This enables probability-based forecasting even when the exact shape of the population distribution is unknown a powerful tool for strategic planning and resource management.

2a. Probability that Sample Mean Revenue Falls Between 3.5 and 3.8 Million INR

Given information:

Population mean: $\mu = 3.7$ million INR

Population standard deviation: $\sigma = 0.52$ million INR

Sample size: $n = 30$

Step 1: Calculate the Standard Error ($\sigma_{\bar{x}}$)

$$\sigma_{\bar{x}} = \sigma / \sqrt{n} = 0.52 / \sqrt{30} = 0.52 / 5.4772 \approx 0.0950 \text{ million INR}$$

Step 2: Convert to Z-scores

For the lower bound ($\bar{X} = 3.5$):

$$Z_1 = (\bar{X} - \mu) / \sigma_{\bar{x}} = (3.5 - 3.7) / 0.0950 = -0.20 / 0.0950 \approx -2.11$$

For the upper bound ($\bar{X} = 3.8$):

$$Z_2 = (3.8 - 3.7) / 0.0950 = 0.10 / 0.0950 \approx 1.05$$

Step 3: Look Up Z-table Probabilities

$P(Z < 1.05) \approx 0.8531$

$P(Z < -2.11) \approx 0.0174$

Step 4: Calculate the Final Probability

$$P(3.5 < \bar{X} < 3.8) = P(Z < 1.05) - P(Z < -2.11) = 0.8531 - 0.0174 = 0.8357$$

Result: The probability that the sample mean daily revenue falls between 3.5 and 3.8 million INR is approximately **0.836, or 83.6%**. This high probability reflects that the CLT produces a tight, normal sampling distribution around the true mean. The standard error of only 0.095 million INR (compared to the original $\sigma = 0.52$) illustrates how averaging across 30 days substantially reduces variability, consistent with the Law of Large Numbers.

2b. Probability that Total Revenue of 30 Random Days is Less than 110 Million INR

The **CLT for Sums** states that for a random sample of size *n*, the distribution of the total (sum) ΣX is approximately normal with:

$$\text{Mean of Sum: } \mu_{\Sigma X} = n \cdot \mu$$

$$\text{Standard Deviation of Sum: } \sigma_{\Sigma X} = \sqrt{n} \cdot \sigma$$

Given information:

$\mu = 3.7$ million INR, $\sigma = 0.52$ million INR, $n = 30$

Target total = 110 million INR

Step 1: Calculate Mean and Standard Deviation of the Sum

$$\mu_{\Sigma X} = 30 \times 3.7 = 111 \text{ million INR}$$

$$\sigma_{\Sigma X} = \sqrt{30} \times 0.52 = 5.4772 \times 0.52 \approx 2.8481 \text{ million INR}$$

Step 2: Calculate the Z-score for $\Sigma X = 110$

$$Z = (\Sigma X - \mu_{\Sigma X}) / \sigma_{\Sigma X} = (110 - 111) / 2.8481 = -1 / 2.8481 \approx -0.35$$

Step 3: Look Up Z-table Probability

$$P(Z < -0.35) \approx 0.3632$$

Result: The probability that the total revenue from 30 randomly selected days is less than 110 million INR is approximately **0.363, or 36.3%**. The theoretical mean sum is 111 million INR, and 110 million INR lies slightly below this value. Since the Z-score is only −0.35, there is a moderate probability of observing totals this low, meaning such a month would not be unusual, but it would represent below-average aggregate performance. This analysis allows city planners to identify months at risk of lower revenue and prepare resource allocations accordingly.

Conclusion

The Central Limit Theorem provides a rigorous statistical foundation for analyzing Pushkar's tourism revenue. By, establishing that the sampling distribution of the mean is approximately normal for $n = 30$ regardless of the underlying population distribution, the CLT enables precise probability-based forecasts. The Law of Large Numbers further guarantees that sample estimates converge to the true population values as sample sizes grow. In this case study, the probability of observing an average daily revenue between 3.5 and 3.8 million INR is very high (83.6%), confirming that Pushkar's revenue is reliably centered around its historical mean. The analysis of total revenue demonstrates that aggregate outcomes follow predictable patterns, which supports informed decision-making in tourism economics and municipal financial planning.

References

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